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A 2-BUTTON SELECTOR For 16 Appliances

Need to control multiple appliances but avoid having a switch for each? Here's a 'low' complexity circuit which will take about '6 hours' to build and cost you less than ₹1100. You will end up with a 2-push button controller for managing up to 16 appliances!

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How do you choose one appliance out of many without a separate switch for each? This project does it with just two push buttons: one to step up through the channels and another to step down. A CD4029 up/down counter keeps track of the selected channel, and a 74LS154 4-to-16 line decoder turns the count into one active output, with four light-emitting diodes (LEDs) showing the selected channel in binary. A CD4093 quad NAND Schmitt-trigger IC gives clean button pulses, while BC557 transistors drive 16 individual 5V relays, so the chosen 230V appliance switches on with the low-voltage electronics safely isolated from the mains. It uses only common logic ICs and no microcontroller, which makes it a compact, low-cost introduction to counters, decoders, and relay switching. Fig. 2 shows the author's breadboard prototype.

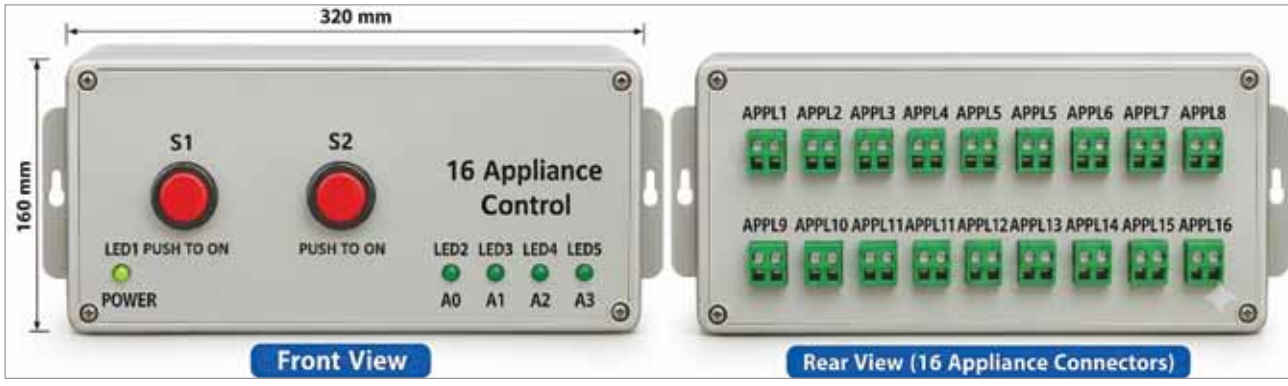


Fig. 1: AI-generated picture of the enclosure

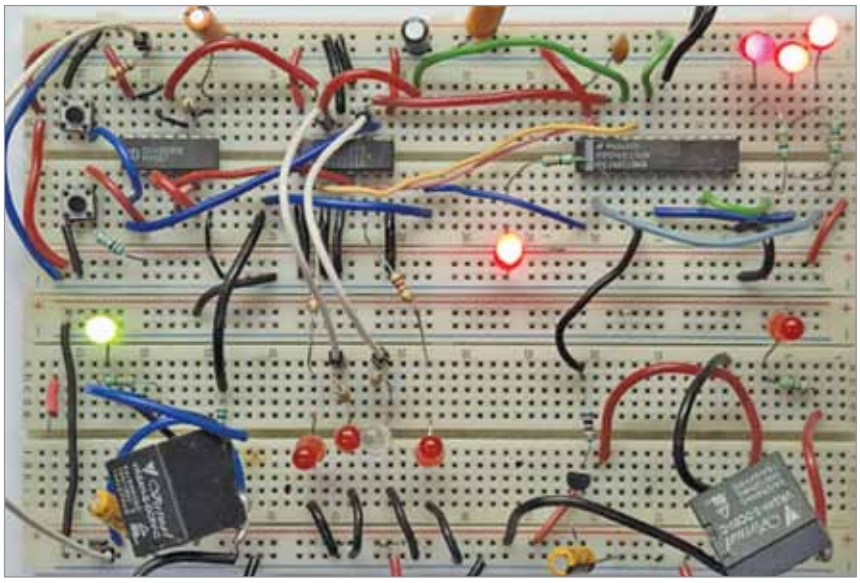


Fig. 2: Author's breadboard prototype

What will you learn?

- How to build a multi-channel digital control circuit using logic ICs, push buttons, counters, decoders, transistors, and relays.
- How a Schmitt-trigger NAND gate removes switch bounce and gives one clean pulse per button press.
- How a 4-bit binary count is converted into 16 individual outputs by a decoder.
- How active-low outputs drive positive-negative-positive (PNP) transistors to energise relays.
- How a 5V electronic circuit can safely switch 230V AC appliances through relay isolation.